

PhacoGuard

An AI second pair of eyes for complex cataract surgery

Research prototype for surgeon awareness during phacoemulsification. Locally hosted models; no clinical data leaves the hospital.

The problem: when a routine cataract turns complex

- Cataract is the leading cause of blindness worldwide^[1]; cataract surgery exceeds 26 million procedures a year^[2].
- Posterior capsule rupture (PCR) is the most feared complication: vitreous loss, dropped nucleus, second surgery, worse vision^[3].
- National registries report PCR in about 1 to 2% of cases (UK Cataract National Dataset 1.92%; NOD 2021-22 0.87%)^{[3][4]}.
- Risk concentrates in complex eyes: weak zonules, dense nuclei, small pupils^[3].
- The warning signs appear seconds to minutes before rupture, and today only the surgeon's divided attention watches for them.

>26 million^[2]

cataract operations every year worldwide

1.92%^[3]

PCR or vitreous loss across 55,567 UK operations

2.8%^[5]

PCR across 3,377 eyes at a Saudi tertiary eye hospital (2019)

1. Cicinelli MV, et al. Cataracts. Lancet. 2023;401:377-89.

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Why Saudi Arabia: a high-complexity case mix

28%^[6]

pseudoexfoliation among cataract candidates aged 60+ in Najran (192/681)

3.5%^[7]

pseudoexfoliation in Riyadh primary-care eye patients (69/1,967), rising steeply after 50

OR 2.14^[8]

pooled odds of PCR or zonular dialysis with pseudoexfoliation (meta-analysis, 22 studies)

OR 2.44^[5]

adjusted odds of PCR with surgical complexity at KKESH; surgeon level not independently associated

- Pseudoexfoliation weakens the zonules and limits pupil dilation, so the lens wobbles and the capsule tears more easily^[8].
- Dense brown nuclei from late presentation need long ultrasound time; brunescent or white cataract is an independent PCR risk factor^[3].
- KKESH data show that complexity, not inexperience, drives PCR, which is exactly what a real-time complexity signal addresses^[5].
- To our knowledge, no AI system for cataract surgical video has been developed in Saudi Arabia or the Gulf; regional deployments are retinal screening tools, not surgical systems.

3. Narendran N, et al. Cataract National Dataset audit of 55 567 operations: PCR and vitreous loss risk. Eye. 2009;23:31-7.

5. Fairaq R, et al. Surgical complexity and surgeon experience in phacoemulsification outcomes: Saudi tertiary eye hospital. Front Ophthalmol. 2026;6:1882657.

6. Gabr AF. Pseudoexfoliation cataract: possible systemic associations. Delta J Ophthalmol. 2016;17:133-6.

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What exists today: research and market

Prior systems guide routine cases or detect complications after the fact; none targets complex-case risk live

System	What it does	Real-time	Complex-case markers	Live risk band	End-of-case record	Saudi or Gulf data
UIC intraoperative guidance, JAMA Ophthalmol 2022 [10]	Pupil tracking, phase recognition, safe-zone overlay; Dice 90%, 97 fps	Yes	No	No	No	No
MetaS, Zhongshan, npj Digit Med 2025 [11]	Capsulorhexis path guidance from 17,538 videos; ideal rhexis rate 16.7% to 64.1%	Yes	Authors flag as future work	No	No	No
CataractCompDetect, arXiv 2025 [12] (preprint)	Detects iris prolapse, PCR, vitreous loss after the fact; PCR F1 0.61	Retrospective	No	No	No	No
Tongren/NUH radial-fold study, medRxiv 2026 [9] (preprint)	Task-specific detector for anterior capsule radial folds; F1 0.87, AUROC 0.96	Clip-level	One marker	No	No	No
Tsukazaki real-time problem detection, J Clin Med 2020 [13]	Real-time phase extraction and instrument tracking with problem alerts	Yes	No	No	No	No
Commercial planning overlays (Zeiss Callisto eye, Alcon Verion)	Toric alignment and incision marks from a pre-operative reference image; no learned intraoperative warnings (manufacturer information)	Overlay	No	No	No	No
PhacoGuard (this proposal)	Three complex-case markers into a calibrated live risk band, plus a grounded end-of-case complication-risk record; fully local stack	Yes	Three	Yes	Yes	Validation on MNGHA data

9. Zhang Y, et al. Task-specific vs general-purpose AI for subtle intraoperative warning signs in cataract surgery. medRxiv [Preprint]. 2026.

10. Garcia Nespola R, et al. AI-based intraoperative guidance tools for phacoemulsification. JAMA Ophthalmol. 2022;140:170-7.

11. Chen X, et al. Digitalization of surgical features improves surgical accuracy (MetaS). NPJ Digit Med. 2025;8:497.

12. Sachdeva B, et al. CataractCompDetect: intraoperative complication detection in cataract surgery. arXiv [Preprint]. 2025.

13. Morita S, et al. Real-time surgical problem detection and instrument tracking in cataract surgery. J Clin Med. 2020;9:3896.

The solution: three markers, one risk band, one record



Marker 1: pupil constriction

Normalised pupil area tracked every frame; a fall during phaco or cortex removal is a recognised PCR risk indicator.^[3]



Marker 2: capsule radial folds

Clip-level detector during capsulorhexis; radial folds are the early visible sign of weak zonules.^[9]



Marker 3: prolonged phaco time

Running phaco duration against the expected distribution; the fingerprint of a dense nucleus.^[3]

Case risk band on a second monitor

Routine

Rising

High

Awareness signal only, never an instruction to the surgeon. Alerts are rate-limited so nothing repeats within 30 seconds.

End-of-case complication-risk record

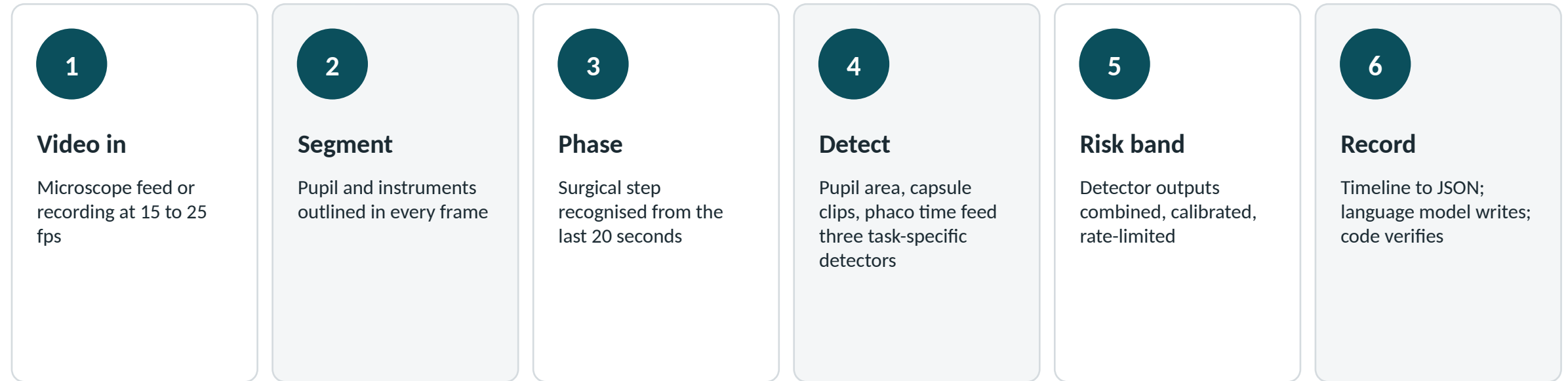
- A locally hosted open-weight language model writes from the structured event timeline only.
- Every number in the record is verified against the timeline by code.
- Output: observed markers, risk tier, suggested follow-up intensity (earlier pressure and inflammation review for high-risk eyes).

3. Narendran N, et al. Cataract National Dataset audit of 55 567 operations: PCR and vitreous loss risk. Eye. 2009;23:31-7.

9. Zhang Y, et al. Task-specific vs general-purpose AI for subtle intraoperative warning signs in cataract surgery. medRxiv [Preprint]. 2026.

How it works

Six steps from microscope video to a verified record, all offline



- Entire stack is open-weight and permissively licensed (Apache 2.0, MIT, BSD) and runs with the network switched off; no clinical data touches an external API, as required by the hackathon rules.
- Task-specific detectors feed the language model rather than asking a general vision model to watch surgery; general-purpose models were unreliable on subtle intraoperative signs (best F1 0.52)^[9].
- Output is observation and a risk state, never surgical guidance (hackathon rule 5.3)^[22].

9. Zhang Y, et al. Task-specific vs general-purpose AI for subtle intraoperative warning signs in cataract surgery. medRxiv [Preprint]. 2026.

22. MNGHA. BeyondVision Hackathon: rules and conditions. 2026.

Evidence, data, and validation plan

Training data: public, licensed, expert-labelled

- Cataract-1K: 1,000 videos, phase labels, pixel masks, pupil-reaction subset; CC BY 4.0^[14]
- CaDIS: 4,670 masked frames including pupil and iris^[15]
- Cataract-101: 101 phase-labelled videos, held out as the external test set^[16]
- Cataract-LMM: about 3,000 videos with phase and segmentation layers^[17]
- Tongren-Zonular-Video: 537 capsulorhexis clips labelled for radial folds^[18]
- MNGHA data: adaptation and validation only, inside the sanctioned environment^[22]

Targets set against published benchmarks

Component	Our target	Published benchmark
Phase recognition	about 90% accuracy	AUROC >0.95 on Cataract-101 [10]; 85 to 96% across architectures [14]
Pupil segmentation	Dice 88 to 90%	Dice 90.2% local, 85.4% external [10]
Radial-fold detection	AUROC about 0.9	F1 0.87, AUROC 0.96, task-specific model [9]
Real-time speed	15+ fps on a laptop GPU	97 fps reported with a dedicated system [10]

Metrics reported: per-marker precision, recall, F1, lead time in seconds, false alerts per case, record factual accuracy by field matching, latency per frame. Video-level splits, never frame-level, to prevent leakage.

Honest scope: detecting rupture itself is hard (published PCR F1 0.61)^[12], so PhacoGuard reports calibrated risk awareness, not diagnosis. Saudi tuning is a design claim until local surgical video is validated; if released on hackathon day, before-and-after results on Saudi eyes will be shown.

9. Zhang Y, et al. Task-specific vs general-purpose AI for subtle intraoperative warning signs in cataract surgery. medRxiv [Preprint]. 2026.
10. García Nespole R, et al. AI-based intraoperative guidance tools for phacoemulsification. JAMA Ophthalmol. 2022;140:170-7.
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15. Grammatikopoulou M, et al. CaDIS: cataract dataset for surgical RGB-image segmentation. Med Image Anal. 2021;71:102053.

16. Schoeffmann K, et al. Cataract-101: video dataset of 101 cataract surgeries. ACM MMSys; 2018.
17. Ahmadi MJ, et al. Cataract-LMM: multi-source multi-task benchmark for surgical video analysis. Sci Data. 2026.
18. Chen L. Tongren-Zonular-Video dataset. Hugging Face; 2026.
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Expected value and compliance

Who benefits

- Surgeons: an objective second observer in the minutes when a case turns complex, with time to deploy a pupil-expansion device or capsular tension ring^[8].
- Patients: fewer ruptures, vitreous losses, and second operations; follow-up intensity matched to the intraoperative risk actually observed.
- MNGHA: a departmental record of complexity and risk events for audit, list planning, and trainee case allocation.
- Measurable: per-marker lead time and false alerts per case; proportion of high-risk records that change follow-up scheduling.

Compliance by design

- Research prototype for surgeon awareness, never guidance (hackathon rule 5.3); no clinical data to external APIs (rule 5.2); institutional data only in the sanctioned environment (rules 4.1, 4.2)^[22].
- Consistent with SFDA MDS-G010, the Kingdom's binding guidance for AI and ML based medical devices^{[19][20]}.
- Aligned with Vision 2030 Health Sector Transformation goals for quality and patient safety^[21].
- Pre-built pipeline declared at registration; adaptation and validation on MNGHA data are the in-event work (rule 3.2). Generative AI use disclosed (rule 5.1).
- Permissive licences throughout: Apache 2.0, MIT, BSD, CC BY 4.0 data.

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21. Saudi Vision 2030. Health Sector Transformation Program.

22. MNGHA. BeyondVision Hackathon: rules and conditions. 2026.

Team and roadmap



2 AI and software engineers

Computer vision, real-time pipelines, on-premise LLM deployment, offline packaging



1 clinical lead

Medical training, MNGHA research access, hardware and software prototyping; owns evaluation design and rules compliance



2 medical experts

Define the risk markers, adjudicate the external test set, validate the clinical framing



MNGHA clinical mentor

Requested for marker validation and hackathon-day adjudication

Roadmap

Now to 12 Oct

Train and validate on public data; package the offline demo; declare pre-built work at registration

13 Oct: hackathon day

Adapt and validate on MNGHA data inside the sanctioned environment; live demonstration; mentor survey

After the hackathon

IRB-approved prospective validation on MNGHA video; departmental pilot; foundation for the first Saudi surgical-video AI dataset; IP review with MNGHA Innovation Division and KAIMRC

References (1 of 2)

Vancouver style. References 9 and 12 are preprints and not yet peer reviewed.

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